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Regular Article

Re-engineering inclusive classroom assessment practices in a digitalized environment. The mediating effect of digital competencies, instructional practices and assessment literacy

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ABSTRACT

Inclusive assessment is crucial for fostering equity and justice in classrooms by accommodating diverse student needs. However, using uniform assessment tools for all students, regardless of their disabilities and learning requirements, has raised significant concerns. This study addresses the gap in research regarding the administration of inclusive assessments by examining how digital competencies, instructional practices, and assessment literacy mediate the relationship between the digital environment and inclusive classroom assessment practices. A cross-sectional survey was conducted with 2317 academic staff members. The study adapted items from previously validated tools and subjected them to rigorous quantitative validation, establishing reliability through Cronbach's alpha. Data analysis included assessing both the measurement model and the structural equation model. The results indicate that the digital environment significantly predicts teachers' practices related to inclusive assessment, digital competence, and assessment literacy. Furthermore, digital competence has a direct impact on instructional practices, assessment literacy, and inclusive assessment. While the joint mediation effects of digital competence and assessment literacy were significant, the combined effect of instructional practices and assessment literacy was not. These findings have important implications for educational policy and practice, highlighting the need for integrating digital competence into teacher training programs to enhance inclusive assessment practices.

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1. Introduction

Inclusive classroom assessment is a vital aspect of modern educational practice, designed to ensure that all students, regardless of their diverse backgrounds and abilities, have fair opportunities to demonstrate their learning. The growing diversity in classrooms, which includes variations in culture, language, learning abilities, and socioeconomic backgrounds, requires the implementation of assessment methods that are equitable, unbiased, and supportive of all students' needs (Edyburn, 2020).

In higher education, classrooms increasingly reflect the diverse society they serve, including students with visual, hearing, physical, and mental impairments. This diversity highlights the urgent need for inclusive classroom assessments that accommodate and support all students, ensuring every learner has a fair opportunity to demonstrate their knowledge and skills. Inclusive classroom assessment has been shown to promote equity and accessibility in education by addressing the various needs of students (Lombardi & Murray, 2011), fostering a supportive learning environment that values diversity by using diverse tools to encourage learning (Thurlow et al., 2011), improving educational outcomes (Bouck, 2020), developing a comprehensive understanding of student capabilities by adapting to diverse learning needs, promoting fairness, enhancing educational experiences, and supporting the academic success of a diverse student population.

Stakeholders have raised concerns about the inclusivity of assessment practices in higher education institutions. Each category of learners requires specific assessment techniques to ensure they can effectively demonstrate their knowledge and skills. Research by supports the use of assistive technologies in assessments to enhance accessibility and performance for visually impaired students. Marschark and Hauser (2011) argue that students with hearing impairments struggle with assessments that are orally or audio-based. Similarly, other scholars have noted that students with physical impairments may not perform as well as their peers in practical physical activities or tasks requiring fine motor skills, and those with mental disabilities may not benefit from assessments that are not tailored to their cognitive processing capabilities (Thurlow et al., 2011).

Previous studies have examined strategies for enhancing inclusive classroom practices in higher education. Early research by Black and Wiliam (1998) indicated that regular, low-stakes assessments provide valuable feedback and allow for timely interventions, benefiting all students, particularly those with learning disabilities. Similarly, advocated for the use of rubrics and formative assessments to cater to different learning styles and abilities. discussed the role of assistive technologies in assessments for visually impaired students, finding that screen readers and braille displays significantly enhance accessibility, enabling these students to participate more fully in assessments. Marschark and Hauser (2011) demonstrated that incorporating these tools improves comprehension and performance, highlighting the need for visual and written formats in assessment practices. Other studies, such as Lombardi and Murray (2011), revealed that adaptive technologies and alternative input methods, like voice recognition software, are critical in creating inclusive assessments for students. Thurlow et al. (2011) investigated assessment practices for students with cognitive disabilities, finding that adaptive testing, which adjusts the difficulty of questions based on the student's responses, is particularly effective in providing an accurate measure of these students' abilities without overwhelming them. This approach aligns with the principles of Universal Design for Learning (UDL), which advocates for flexible assessment methods to meet diverse needs.

Despite extensive research on inclusive assessment practices, these studies have not considered other factors such as the digital environment in relation to inclusive classroom assessments, particularly regarding the mediating roles of instructional practices, assessment literacy, and digital competencies. Additionally, the contexts in which these studies were conducted differ from the locale of this study, making it risky to generalise their findings due to cultural, technological, and diversity differences. The existing literature mainly focuses on traditional assessment methods and their accommodations for various disabilities. Although there is a growing interest in digital assessments, the integration of digital tools with inclusive practices remains underexplored. Specifically, the combined effect of instructional practices, assessment literacy, and digital competencies on the effectiveness of digital inclusive assessments has not been thoroughly investigated. This study aims to address these gaps by exploring the impact of the digital environment on inclusive classroom assessments in higher education, considering the mediating roles of instructional practices, assessment literacy, and digital competencies.

This study has significant theoretical implications, as it seeks to integrate and extend existing theories on inclusive education, digital learning environments, and assessment practices. By examining the intersection of digital environments and inclusive assessments, this research contributes to the body of knowledge on inclusive education. It underscores the need for theoretical frameworks that account for the dynamic interplay between technology and diverse learner needs. The findings can inform theories on Universal Design for Learning (UDL), highlighting the role of digital tools in creating flexible and inclusive assessment methods. The results can guide educators in designing and implementing more inclusive digital assessments. By understanding the mediating roles of instructional practices, assessment literacy, and digital competencies, educators can develop tailored strategies to support diverse learners. This includes providing professional development to enhance teachers' digital competencies and assessment literacy, ensuring they are equipped to create inclusive assessments. The study offers evidence-based recommendations for institutional policies that promote inclusive assessment practices, including investing in digital infrastructure, providing resources for assistive technologies, and establishing guidelines for inclusive digital assessments. Thus, the following research questions were raised to guide the study.

i. What is the direct impact of digital environment on inclusive classroom assessment practices?
ii. How does digital environment impact on digital competence?
iii. What is the impact of digital environment on assessment literacy?
iv. How digital environment impact on instructional practices?
v. To what extent does digital competences impact on instructional practices?
vi. What is the direct impact of digital competences on assessment literacy?
vii. What is the direct impact of digital competence on inclusive classroom practices?
viii. How does digital competences mediate the nexus between digital environment and inclusive classroom assessment practices?
ix. What is the direct impact of instructional practices on assessment literacy?
x. What is the impact of instructional practices on inclusive classroom assessment?
xi. How does instructional practices mediate the linkage between digital competencies and inclusive classroom assessment?
xii. To what extent does instructional practices mediate the nexus between digital environment and inclusive classroom assessment practices?
xiii. What is the direct impact of assessment literacy on inclusive classroom assessment practices?
xiv. How does assessment literacy mediate the linkage between digital competencies and inclusive classroom assessment?
xv. To what extent does assessment literacy mediate the relationship between digital environment and inclusive classroom assessment practices?
xvi. How does instructional practices and assessment literacy mediate the relationship between digital environment and inclusive classroom assessment?

xvii. What is the mediating effect of digital competence and assessment literacy on the relationship between digital environment and inclusive classroom assessment?
xviii. How does digital competence and Instructional practices mediate the relationship between digital environment and inclusive classroom assessment?

2. Literature review and hypothesis development

2.1. Studies on digital environment

The integration of digital environments into education has transformed teaching and assessment practices, creating new opportunities for inclusivity. Digital tools and platforms can be customised to meet diverse learner needs, making assessments more accessible for students with disabilities. For instance, assistive technologies such as screen readers, braille displays, and text-to-speech software enable visually impaired students to participate effectively in digital assessments. Highlighted the importance of these technologies in levelling the playing field for students with visual impairments, allowing them to access and respond to assessment materials independently. Bouck (2020) noted that digital environments allow educators to create personalized assessment experiences that cater to individual student requirements, thereby promoting inclusivity.

In environments where digital tools are effectively utilised, the potential to diversify assessments to meet all students' needs is high. This is not to say there are no challenges; for example, disparities in access (Warschauer, 2010), hardware malfunctions (Mavrou et al., 2017), and user capabilities can affect utilisation. However, the benefits of using these tools in modern assessment outweigh the challenges. Similarly, digital environments provide teachers with tools for effective assessment practices and the competencies required for these practices. For example, teachers can acquire literacy in the use of differentiated assessment tools through online materials that enhances their professional skills. Therefore, it is imperative to examine this nexus for policy purposes.

Previous studies have explored the digital environment in educational settings. For instance, Smith et al. (2018) found that digitalising the school environment increased student participation and engagement in classroom activities. Johnson and Brown (2020) also found that students who used digital platforms performed better in exams due to the interactive and inclusive features provided. Other researchers support these findings (Garnett et al., 2021; Huang & Hong, 2019; Lee & Kim, 2018; Martinez & Perez, 2020; Rodriguez & Garcia, 2020). These findings highlight the need for digital environments to promote performance and personalise experiences. However, these studies did not specifically capture the relationship between the digital environment and inclusive classroom assessment, which is the focus of this study. It is imperative that in digital environments created for inclusivity, teachers' ability to diversify assessments to meet learners' needs will be high, and assessments will be highly differentiated to reduce systematic errors previously present in classroom assessments. Thus, the first hypothesis was stated.

Hypothesis 1. Digital environment has no significant impact on inclusive classroom assessment practices.

Hypothesis 2. Digital environment has no significant impact on digital competence.

Hypothesis 3. Digital environment has no significant impact on assessment literacy.

Hypothesis 4. Digital environment has no significant impact on instructional practices.

2.2. Studies on digital competence

Digital competence refers to the ability to use digital technologies effectively, safely, and critically in various contexts. In education, digital competence encompasses a range of skills, including proficiency in using digital platforms, the ability to locate and use digital information, communicate and collaborate with others, and create content, among other skills (Ferrari, 2013). Inclusive assessment, which involves designing assessments that cater to the needs of all students, will benefit greatly if teachers are competent in handling digital tools. Integrating digital competence with inclusive assessment practices enhances the ability to create a supportive and equitable learning environment for all students. This integration allows educators to leverage digital tools to design and implement assessments that cater to diverse learning needs (Thurlow et al., 2013). For instance, technologies such as screen readers, speech-to-text software, and digital note-taking tools support students with disabilities in accessing and completing assessments. A student with a visual impairment, for example, can use a screen reader to navigate digital assessments and complete tasks independently (Raskind & Margalit, 2021). Additionally, students can use tools like Kahoot! Quizlet, and Socrative to engage through interactive quizzes and games. These platforms can be customised to provide various question types and formats, catering to different learning styles and abilities (Alamri & Tyler-Wood, 2017; Castellano, 2017; Edyburn, 2020).

The study by Alamri and Tyler-Wood (2017) revealed that teachers who improved their digital competencies were better able to design and implement inclusive assessments that catered to the diverse needs of their students. Castellano (2017) found that these tools provided more flexibility and accessibility for students with disabilities, allowing them to demonstrate their knowledge and skills in multiple ways. Moreover, the use of digital tools led to increased student motivation and participation in assessments. Houghton and Smith (2023)findings reveal that higher levels of digital competence are positively associated with more inclusive assessment practices. Johnson and Williams (2023)study surveyed university instructors and analyzed how their proficiency with digital tools affects their approach to inclusive assessment. Results indicate that educators with advanced digital skills are better equipped to create assessments that are accessible and fair to all students. Lee and Martin. (2023) study examines how digital competence among academic staff contributes to inclusive assessment practices in higher education institutions. The researchers found that high levels of digital competence are correlated with more effective and inclusive assessment practices. Pate and Nguyen, (2023)study uses quantitative analysis to assess the impact of digital skills training on the ability of educators to implement inclusive assessment practices. Findings indicate that digital competence significantly enhances the ability to design and conduct assessments that address the diverse needs of students. Other studies have shown that teachers' digital competence is essential in maintaining effective assessment practices in inclusive settings (Edyburn, 2020; Meyer et al., 2014; Raskind & Margalit, 2021). These studies highlight the importance of leveraging digital technologies to create accessible, flexible, and personalized assessment methods that cater to the diverse needs of all students, particularly those with disabilities. However, it is not clear whether these studies have been conducted using a complex model like the one in this study. This is because what holds true as a single factor may function differently when considered alongside other factors, especially in behavioural science where behaviours are influenced by numerous variables. Thus, the following hypotheses were stated.

Hypothesis 5. Digital competences has no significant impact on instructional practices

Hypothesis 6. Digital competences has no significant impact on assessment literacy.

Hypothesis 7. Digital competences has no significant impact on

inclusive classroom practices.

Hypothesis 8. digital competences do not mediate the nexus between digital environment and inclusive classroom assessment practices.

2.3. Studies on instructional practices

Instructional practices refer to the strategies and methods employed by educators to facilitate learning and ensure that students acquire the necessary knowledge and skills. Effective instructional practices are characterised by their ability to engage students, promote critical thinking, and adapt to the diverse needs of learners (Tomlinson, 2014). Effective instructional practices can incorporate elements of differentiated instruction, which accommodates varying abilities, learning styles, and interests of students (Tomlinson, 2014), active learning that enhances students' understanding and retention of information (Prince, 2004), and the use of technology that promotes interactive discussions (Hattie, 2009). For example, in a science class, a teacher might use differentiated instruction to present content through videos, hands-on experiments, and textual resources. Concurrently, they can employ inclusive assessment practices by offering students the choice to demonstrate their understanding through written reports, oral presentations, or digital projects. This approach ensures that all students, regardless of their abilities or learning preferences, can engage with the material and showcase their knowledge.

The connection between instructional practices and inclusive assessment practices is crucial for providing equitable educational opportunities to all students. Current research emphasizes that effective teaching methods are essential for creating assessment environments that accommodate diverse learner needs. For example, Lee and Martin (2023) show that instructional strategies focusing on differentiation and adaptive learning significantly enhance inclusive assessment practices. Their findings indicate that when educators use various pedagogical approaches tailored to individual learning styles, it increases the fairness and accessibility of assessments. Likewise, Patel and Nguyen (2023) report that inclusive instructional methods, such as collaborative and project-based learning, aid in developing assessments that consider diverse perspectives and abilities, thus fostering a more inclusive educational environment. These insights highlight the importance of combining inclusive teaching strategies with assessment design to ensure fair and effective evaluation of all students, promoting a comprehensive understanding of their abilities and progress (Houghton & Smith, 2023; Johnson & Williams, 2023).

Several studies have examined instructional practices, particularly the efficacy of differentiated instructional techniques in educational settings. For instance, Tomlinson and Moon (2013) explored the impact of differentiated instruction on student achievement in inclusive classrooms. The results showed a significant improvement in academic performance across all student groups, with students with learning disabilities showing the most considerable gains. In another study, Black and Wiliam (1998) found that formative assessment significantly increased student engagement and motivation, leading to higher academic achievement. Other studies have also demonstrated the effectiveness of differentiated instructional practices. Meyer et al. (2014) found that students with disabilities showed improved academic performance and greater participation in classroom activities when a Universal Design for Learning approach was used. However, the context of these studies differs. It is not known to the researchers if these practices are common in the context of this study. Hence, the following hypotheses were formed.

Hypothesis 9. Instructional practices have no significant impact on assessment literacy.

Hypothesis 10. Instructional practices have no significant impact on inclusive classroom assessment.

Hypothesis 11. Instructional practices do not mediate the linkage between digital competencies and inclusive classroom assessment.

Hypothesis 12. instructional practices do not mediate the nexus between digital environment and inclusive classroom assessment practices.

2.4. Studies on assessment literacy

Assessment literacy refers to the knowledge and skills required by educators to develop, implement, and interpret assessments effectively. This includes understanding various assessment types, designing fair and reliable assessments, and using data to inform instruction and improve student learning. For educators to create an inclusive classroom environment, a high level of assessment literacy is crucial. For instance, teachers must understand various assessment tools (Black & Wiliam, 1998), design fair assessments (Ofem et al., 2023), interpret data (Popham, 2011), and provide feedback (Hattie & Timperley, 2007). This knowledge is important because assessments developed in an inclusive setting must accommodate all students, allowing for alternatives and adjustments (Wiliam, 2011). Indeed, these types of assessments have posed challenges for many academics in higher education. Integrating assessment literacy with inclusive assessment practices ensures that educators can effectively support all students. By understanding and implementing inclusive strategies, teachers can design assessments that not only measure student learning accurately but also promote equity and accessibility. However, extensive surveys have not been conducted to explore this nexus. This highlights the need to understand these implications to provide sufficient empirical evidence. Hence, the following hypotheses were stated.

Hypothesis 13. Assessment literacy has no direct effect on inclusive classroom assessment practices.

Hypothesis 14. Assessment literacy do not mediate the linkage between digital competencies and inclusive classroom assessment.

Hypothesis 15. Assessment literacy do not mediate the relationship between digital environment and inclusive classroom assessment practices.

2.5. Studies on inclusive classroom assessment practices

Inclusive classroom assessment practices are vital for ensuring that all students, irrespective of their abilities or backgrounds, have equitable opportunities to demonstrate their learning. These practices involve adapting assessment methods to accommodate diverse learners, including those with disabilities, and implementing strategies that promote fairness and accessibility. There is documentary evidence on inclusive classroom assessment practices. For instance, Olufemi and Adebayo (2018) explored the implementation of inclusive education and assessment practices in Nigerian primary schools. Their findings indicated that while there is awareness of inclusive assessment practices, many teachers lack the necessary training and resources to implement them effectively. Aderemi and Olaleye (2019) found that although teachers generally support the idea of inclusive assessments, they face significant barriers such as large class sizes, insufficient training, and lack of access to assistive technologies. In Africa, Donohue and Bornman (2018) found that despite a strong commitment to inclusive education, schools often struggle with limited resources and training. Similarly, Mukuria and Korir (2019) identified a lack of training and resources as major barriers to effective inclusive assessment. This contrasts with Europe, where significant progress has been recorded in inclusive classroom assessment practices. Saloviita (2020) found that Finnish schools employ a range of inclusive assessment strategies, including differentiated instruction, formative assessments, and the use of digital tools. Other studies have shown that inclusive assessment practices have advanced considerably in Europe (Norwich & Lewis, 2017; Rodríguez-Muñiz et al., 2019).

The reviewed studies demonstrate a growing recognition of the importance of inclusive assessment practices in diverse educational contexts. However, there are notable gaps and challenges that need to be addressed to improve these practices, especially in Nigeria and Africa, where issues of resource allocation, professional development, and policy support have been significant obstacles. It is imperative that this study be carried out to provide empirical evidence that will guide policy development and offer a nuanced understanding of the complex factors that influence inclusive assessment practices. Thus, the following hypotheses were formulated.

Hypothesis 16. Instructional practices and assessment literacy do not mediate the relationship between digital environment and inclusive classroom assessment.

Hypothesis 17. digital competence and assessment literacy do not mediate the relationship between digital environment and inclusive classroom assessment.

Hypothesis 18. digital competence and Instructional practices do not mediate the relationship between digital environment and inclusive classroom assessment.

2.6. Conceptual framework

The conceptual framework delves into the complex interactions between the digital environment, digital competence, instructional methods, assessment literacy, and inclusive assessment practices in education. It posits that a well-established digital environment fosters digital competence, which in turn influences instructional methods and assessment literacy. These factors together impact the effectiveness of inclusive assessment practices, ensuring that evaluations are fair and cater to a wide range of learner needs.

The digital environment, encompassing technological infrastructure and resources, facilitates the incorporation of technology into teaching, thereby enhancing educators' digital skills. Davis's Technology Acceptance Model (1989) highlights that the perceived ease of use and usefulness of technology play a crucial role in its adoption, thereby improving digital competence. Educators with strong digital skills are better equipped to implement varied instructional approaches and assessment methods, thereby boosting assessment literacy. Venkatesh et al.'s Unified Theory of Acceptance and Use of Technology (2003) elaborates on how factors like performance expectancy and social influence affect technology usage, which in turn shapes instructional methods and assessment literacy. Effective teaching practices and high assessment literacy contribute to inclusive assessments, ensuring equitable evaluations for all students, as supported by constructivist theories and the UTAUT model. The framework for this study was constructed based on the researchers' theoretical understanding of these variables, as depicted in Fig. 1.

3. Methodology

The study utilised a cross-sectional design, which was appropriate as it involved collecting data from respondents engaged in various activities simultaneously without manipulating the variables. The research primarily examined the relationships between the digital environment, digital competence, instructional practices, assessment literacy, and inclusive classroom assessment practices. The participants were academic staff in tertiary institutions. The study sampled 2317 academic staff, representing 10% of the 23,170 staff from six public universities in three geopolitical zones in Nigeria. The sampling was done in clusters. In each cluster (university), 10% of the staff were included in the study. Descriptive statistics of the respondents showed that 1421 (61.33%) were male and 896 (38.67%) were female. Additionally, 679 (29.30%) were under 35 years old, 1216 (52.48%) were between 35 and 50 years old, and 422 (18.21%) were above 50 years old. Among the participants, 378 (16.31%) held the rank of assistant lecturer to lecturer II, 1416 (61.11%) were within the rank of lecturer I to senior lecturer, and 523 (22.57%) were professors.

3.1. Instruments and measures

Five measures were utilised in this study: digital environment, digital competencies, instructional practices, assessment literacy, and inclusive classroom assessment practices. The instrument was divided into three sections: A, B, and C. Section A collected demographic information, such as age, gender, and professional rank, using a categorical scale. Section B gathered information about the independent variables, mediators, and dependent variables on a continuous scale. Following an extensive literature review, scales measuring the constructs of interest were identified, and the items in these scales were adapted to suit the context and purpose of the study. A digital school environment, operationalised as an environment where technology is integrated into all aspects of the educational experience—including teaching, learning, and administration—to enhance student engagement and learning outcomes, was measured with items adapted from the original scale of Tondeur et al.

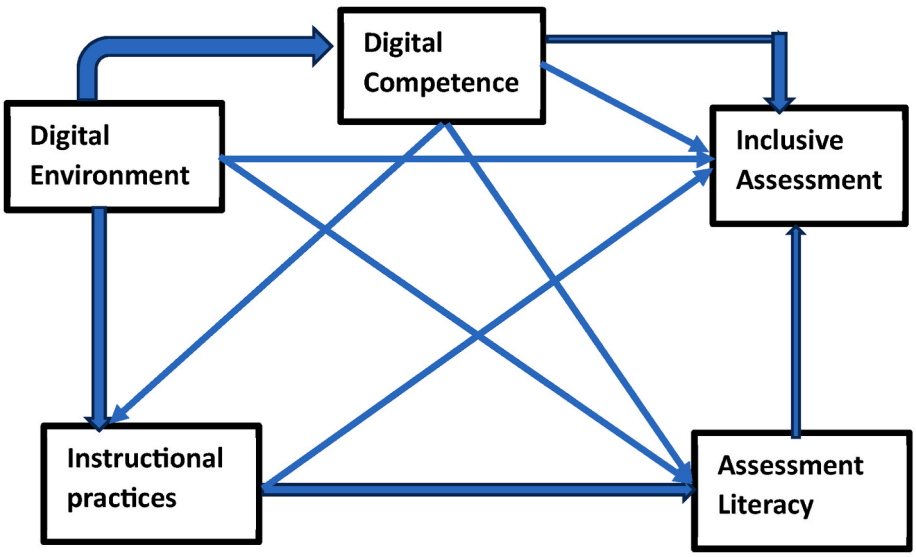


Fig. 1. Conceptual framework of the nexus between digital environment and inclusive classroom assessment practices in the presence of three mediators.

(2017). The scale has a total of 24 items and eight items adapted where restructured based on the context of the study with a sample item: "My school has digital facilities that are used for teaching and learning." Digital competence, referring to the skills and knowledge required to effectively use digital tools and resources in teaching practices, was measured with items adapted from the original scale of Spiteri and Rundgren (2020). The scale has a number of 22 items measuring diverse dimensions with key emphasis on competence. A total of eight items were selected to suit the purpose of the study with sample item: "I integrate digital resources into my lessons to enhance student learning." Instructional practices, encompassing the methods and strategies used to facilitate learning, were measured with items adapted from the scale developed by Sawada et al. (2002). The original scale has 25 items bothering on six areas but eight items were adapted to suit the context of the study with a sample item: "I ensure that methods used for instruction cater to all students' needs." Assessment literacy items were adapted from the scale of Mertler and Campbell (2005), with a total of 35 items. In this study a total of eight items were sected according to the purpose of this study with a sample item being: "I understand how to develop assessment tasks that align with students' peculiarities." All items for the independent and mediator variableswere reported to have adequate psychometric properties. Section C comprised items measuring inclusive classroom assessment practices. This construct was measured with 10 items adapted from the Inclusive Teaching Strategies Inventory (ITSI) developed by Lombardi et al. (2011). Two sample items, modified to suit the study, were: "I provide multiple ways for students to demonstrate their understanding" and "I make adjustments in my assessment plans to accommodate students with disabilities." All items were assessed using a four-point scale ranging from 1 (strongly disagree) to 4 (strongly agree) and were calibrated to suit academic staff in tertiary institutions within the context of this study.

3.2. Content validity

The scale items underwent quantitative validation by five experts representing various fields. These experts evaluated the relevance, clarity, and suitability of the items across the study's different constructs. Initially, 42 items were developed based on a thorough literature review and adaptation from prior studies. Following initial screening, two items (ICA8 and ASS1) were excluded. Experts received a questionnaire and an assessment rubric, following guidelines proposed by Ofem et al. (2024a) Ofem, Idika, et al., 2024. Each item was rated on a scale of 1–4, with 1 indicating irrelevance and 4 indicating high relevance. The Item-Content Validity Index (I-CVI), which ranges from 0 to 1, was then calculated. A threshold of 0.78 or higher is commonly accepted, indicating agreement among at least 78% of experts regarding an item's relevance or clarity (Haynes et al., 1995; Anastasi & Urbina, 1997). However, some researchers consider values as low as 0.70 acceptable under certain conditions.

For the Scale-Level Content Validity Index (S-CVI), typically assessed via S-CVI/Ave (average method) and S-CVI/UA (universal agreement method), a value of 0.80 or higher is generally accepted, indicating that, on average, at least 80% of items are deemed relevant or clear by the expert panel (DeVellis, 2016; Polit & Beck, 2006). In this study, I-CVI values for the three criteria ranged from 0.80 to 0.89, while S-CVI values ranged from 0.90 to 0.97, indicating a high level of relevance and clarity as assessed by the expert panel. Subsequently, four items (DC1, INS1, ASS3, ASS6) were deemed unfit and removed, resulting in a total of 36 items for use in the study. Reliability testing of the instrument involved 100 staff members not involved in the study, showing high internal consistency across sub-scales.

3.3. Procedure for data collection

Data collection strictly adhered to ethical research guidelines, including guidelines from the Nigeria Code for Health Research Ethics. The data collection process for this study was carefully designed to ensure accuracy, confidentiality, and replicability. Ten research assistants were chosen from the participating public universities to act as the primary contacts within their institutions. These assistants, who were financially compensated, underwent thorough training to ensure they understood the data collection procedures and could effectively communicate the study's objectives to participants.

Prior to distributing the questionnaires, academic staff were fully briefed on the purpose of the study. They were assured that their responses would remain confidential, with no third-party access, and informed that the data collected would contribute to research findings to be published in an international journal. This transparency helped to build trust and foster participation.

The research assistants manually distributed the questionnaires to 2317 selected academic staff members, which represented 10% of the total staff population of 23,170 from six public universities across three geopolitical zones in Nigeria. Of the 2317 questionnaires distributed, 342 academic staff opted out for personal reasons and were excluded from the study. As a result, 1975 staff responded, leading to an 85% response rate. Out of these, 1954 questionnaires were properly completed and deemed suitable for analysis using Smart PLS 3.

4. Result

4.1. Measurement model and quality assessment criteria

The model was evaluated to assess the quality of item loadings for each factor. Research suggests that item loadings above 0.70 are preferable (Memon & Rahman, 2014; Ofem et al., 2024d). Table 1 indicates that all item loadings exceed 0.70, except for DC5, ICA3, INS1, INS2, and

Table 1
Quality criteria assessment of the constructs underlying the study.

Items	α	CR	AVE	EFA	VIF	R^2	F^2
Assessment literacy	0.876	0.915	0.729			0.203	0.048
ASS1				0.857	2.28		
ASS2				0.848	2.297		
ASS3				0.845	2.188		
ASS5				0.865	2.313		
Digital competence	0.829	0.874	0.595			0.139	0.068
DC1				0.825	1.978		
DC2				0.917	3.290		
DC3				0.785	1.957		
DC4				0.850	2.245		
DC5				0.362	1.208		
Digital environment	0.871	0.912	0.722				
DE1				0.894	3.051		
DE2				0.803	2.275		
DE3				0.856	2.626		
DE6				0.844	2.410		
Inclusive assessment	0.877	0.911	0.676			0.433	0.178
ICA1				0.896	3.493		
ICA2				0.795	2.310		
ICA3				0.635	1.471		
ICA4				0.897	3.790		
ICA5				0.859	3.074		
Instructional practice	0.813	0.871	0.580			0.033	0.092
INS1				0.612	1.379		
INS2				0.673	1.430		
INS3				0.942	5.205		
INS4				0.675	1.517		
INS5				0.856	3.757		

EFA = Exploratory factor analysis, VIF=Variance Inflation Factor; CR=Composite Reliability, AVE = Average Variance Extracted; α = Cronbach alpha, ASS=Assessment literacy, DC = Digital competence, DE = Digital environment, ICA=Inclusive classroom assessment, INS=Instructional practices.

INS4. Experts suggest that items below 0.70 can still be retained if the overall construct shows validity and reliability (Hair et al., 2019) and meets content validity criteria (Diamantopoulos et al., 2012; Ofem et al., 2024b). Research supports that minor deviations are acceptable in exploratory research if the overall model fit is maintained (Henseler et al., 2009). Theoretical justifications for retaining such items can help maintain conceptual integrity (Gefen et al., 2000). However, future research should either revise or improve these items due to their poor loading on the overall construct.

At the scale level, Cronbach's alpha values for all constructs exceed 0.70, indicating internal consistency. Convergent validity was assessed using the average variance extracted (AVE), with a threshold of 0.50 or higher indicating sufficient evidence. Table 1 demonstrates that all variables achieved convergent validity. Discriminant validity was evaluated to ensure that unrelated variables were not highly correlated. The Fornell–Larcker criterion (Fornell & Larcker, 1981) was applied, comparing the square root of the AVE with the correlation of a factor with other factors in the model (Ofem, Idika, et al., 2024; Ofem, Idika, et al., 2024). Table 2 illustrates that discriminant validity was achieved for the three constructs, as the diagonal values (square root of the AVE) exceed the correlation coefficients below them.

To assess the predictive relevance of the model, potential collinearity within the structural model was initially identified. Table 1 indicates that the outer variance inflation factors (VIFs) for all constructs did not surpass the recommended threshold of 5.00 (Hair et al., 2017), ranging from 1.54 to 5.00. Similarly, the VIFs for the inner model were below 5.00, ranging from 1.00 to 1.05 (see Table 1), indicating no significant collinearity among the predictor constructs in the structural model.

Subsequently, the proportion of variance in the criterion variables explained by the exogenous variables was examined to evaluate the model's in-sample fit and predictive accuracy. Table 1 reveals that the digital environment accounts for 17.9% ($R^2 = 0.179$) of the variance in inclusive classroom assessment practices. Together, digital environment and digital competence explain 3.3% ($R_2 = 0.033$) of the variance in instructional practices. When considering digital environment, digital competence, and instructional practices together, they explain 20.3% of the variance in assessment literacy among teachers. However, all exogenous variables combined explain 43.3% ($R^2 = 0.434$) of the variance in inclusive classroom assessment practices, leaving 56.7% unexplained. According to established guidelines, predictive accuracy below 10% is considered weak (Hair et al., 2013). Only the joint effect of digital competence and environment on instructional practices falls below this threshold. Nonetheless, the relatively low R^2 value reflects the model's complexity with a limited number of predictors. The model is accepted because behavioral sciences acknowledge that multiple factors contribute to variances in endogenous variables.

4.2. Structural model and test of hypotheses

The hypotheses of this study were tested according to the conceptual model to determine the direct and indirect effects of the variables.

The result for H1 to H4 is presented in Table 3 and Fig. 2. The result for direct effect of digital environment on inclusive assessment practices showed a significant positive direct effect of digital environment on inclusive assessment practice (β = 0.35, 95% CI [0.32,0.39], t = 18.51, p < .001). For H2, which examined the direct effect of digital environment on teachers' digital competences, the result was (β = 0.37, 95% CI [0.37,0.33], t = 20.04, p < .001), indicating that H2 is rejected. Similarly, for H3, which examines the direct effect of digital environment on assessment literacy, the result was (β = 0.28, 95% CI [0.25,0.25], t = 14.04, p < .001). This indicates that digital environment directly affects assessment literacy of teachers. For H4 which measure the impact of digital environment on instructional practices ((β = -0.02, 95% CI [0.2-0.02,.-0.05], t = 1.01, p > .05).The outcome of the four-hypothesis tested is that digital environment directly impacts on teachers' inclusive assessment practices, digital competences, assessment literacy but not instructional practices. Thus, hypothesis for H1, H2, H3, is rejected and H4 is retained.

The result of hypothesis 5 to hypothesis 8 are presented in Tables 3 and 4. The result for digital competence on instructional practices revealed that practice (β = 0.19, 95% CI [0.15,0.22], t = 9.94, p < .001). The result for digital competence and assessment literacy showed that β = 0.25, 95% CI [0.21,0.30], t = 911.16, p < .001). hypothesis 7 which is on the impact of digital competence on inclusive assessment practices (β = 0.22, 95% CI [0.18,0.26], t = 10.96, p < .001). Thus, hypotheses 5,6 and 7 are rejected and alternate hypothesis supported. The result for the mediating effect of digital competencies in the relationship between digital environment and inclusive classroom assessment practices as presented in Table 4 revealed that (β = 0.08, 95% CI [0.06,0.10], t = 9.67, p < .001). Thus, hypothesis 5, 6, 7, 8 are rejected.

The result of hypothesis 9,10,11 and 12 is presented in Tables 3 and 4 The result for instructional practices and assessment literacy revealed that (β = -0.06, 95% CI [-0.09,.-0.03], t = 4.09, p < .001). for H10, which is on instructional practices and inclusive classroom assessment pratices (β = 0.23, 95% CI [0.20,0.26], t = 16.03, p < .001). For H11 which is on the mediating effect of instructional pratices in the nexus between digital competence and inclusive classroom assessment pratices, the result in Table 4 revealed that (β = 0.04, 95% CI [0.03,0.05], t = 8.49, p < .001) and for digital competence and inclusive classroom assessment pratices as mediated by assessment pratices (β = -0.04, 95% CI [0.03,0.06], t = 8.26, p < .001). Thus, the alternate hypothesis for 9.10.11, and 12 are accepted.

The result of hypothesis 13, 14, and 15 is presented in Tables 2 and 3 The result for assessment literacy and inclusive assessment practices revealed that (β = 0.19, 95% CI [0.14, 0.22], t = 9.71, p < .001). for H14, which is on assessment pratices mediating the nexus between digital competence and inclusive classroom assessment pratices (β = 0.04, 95% CI [0.03,0.06], t = 8.26, p < .001). For H15 which is on the mediating effect of instructional pratices in the nexus between digital competence and inclusive classroom assessment pratices, the result in Table 4 revealed that (β = 0.04, 95% CI [0.03,0.05], t = 8.26, p < .001) and for digital environment and inclusive classroom assessment pratices as mediated by assessment pratices (β = -0.05, 95% CI [0.04,0.06], t = 7.66, p < .001). Thus, the alternate hypothesis for 13.14, and 15 are accepted.

The result of hypotheses 16, 17,18 and 19 test the joint mediation of the variables is presented in Tables 2 and 3 The result for joint mediation of instructional pratices and assessment literacy on inclusive assessment practices revealed that (β = 0.00, 95% CI [0.00, 0.01], t = 0.97, p > .05). For H17, which is that there is no joint mediation effect of digital competence and assessment literacy on inclusive assessment practices revealed that (β = 0.01, 95% CI [0.01,0.02], t = 7.62, p < .001). For H18 which is on the joint mediation effect of digital competence and instructional pratices on inclusive assessment practices revealed that (β = 0.01, 95% CI [0.01,0.02], t = 7.84, p < .001) and for H19 which is on the joint mediation effect of instructional pratices, digital competence and assessment literacy on inclusive assessment practices revealed that (β = −0.01, 95% CI [-0.00,0.02], t = 3.13, p < .001). This implies that alternate hypothesis for 17, 18, and 19 were rejected but retained for hypothesis 16.

Table 2
Discriminant validity evidence through the Fornell–Larcker.

Construct	1	2	3	4	5
Assessment Literacy (1)	**0.854**				
Digital competence (2)	0.356	**0.773**			
Digital Environment (3)	0.383	0.373	**0.850**		
Inclusive assessment (4)	0.401	0.464	0.524	**0.822**	
Instructional Practices (5)	−0.003	0.18	0.05	0.291	**0.762**

Note: Square roots of AVE are bolded along the diagonal; Factor correlations are below the leading diagonal.

Table 3
Direct effects.

Linkages	β	M	95%CI		STD	t-val	p-val
ASS - > ICA	0.185	0.185	0.148	0.222	0.019	9.712	0.000
DC - > ASS	0.259	0.26	0.213	0.305	0.023	11.165	0.000
DC - > ICA	0.223	0.223	0.183	0.263	0.02	10.961	0.000
DC - > **INS**	0.187	0.188	0.15	0.223	0.019	9.945	0.000
DE - > ASS	0.289	0.289	0.25	0.329	0.021	14.043	0.000
DE - > DC	0.373	0.373	0.336	0.409	0.019	20.043	0.000
DE - > ICA	0.359	0.359	0.321	0.397	0.019	18.518	0.000
DE - > INS	−0.02	−0.021	−0.057	0.019	0.02	1.018	0.309
INS - > ASS	−0.064	−0.064	−0.095	−0.034	0.016	4.088	0.000
INS - > ICA	0.233	0.233	0.203	0.261	0.015	16.034	0.000

M = mean, STD = standard deviation, DE = digital environment, DC = digital competence, INS = instructional practice, ASS=Assessment literacy, ICA-inclusive assessment practice.

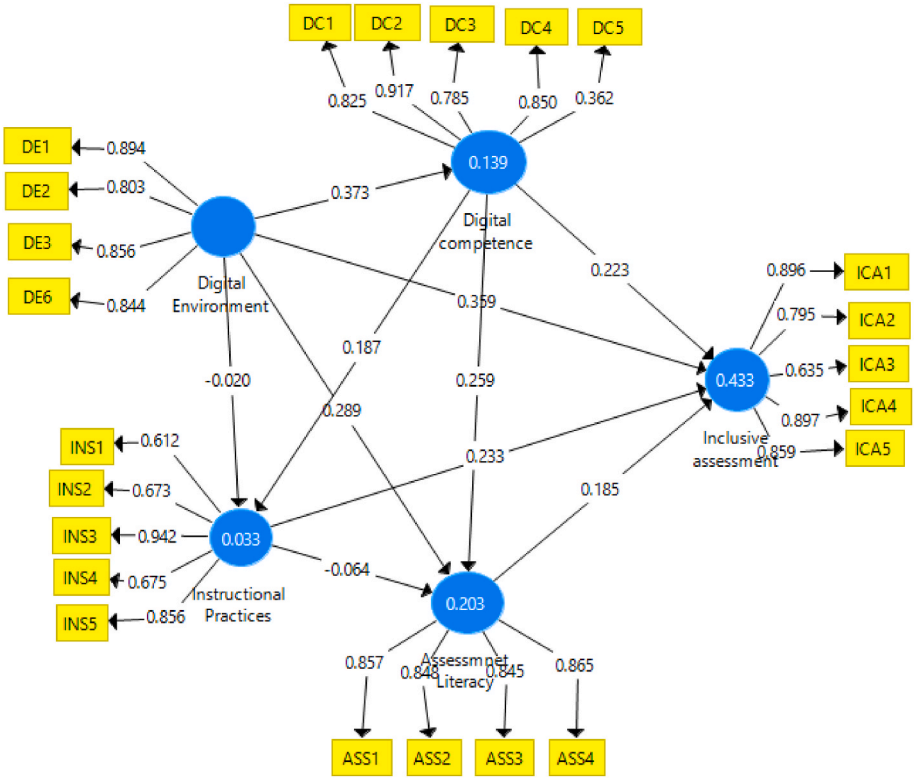


Fig. 2. Structural equation model of the linkage between digital environment as mediated by digital competencies, assessment literacy and instructional pratices on inclusive classroom practices.

Table 4
Indirect effects.

Linkages	β	M	95%CI		STD	t-val	p- val
DE - > DC - > ASS	0.097	0.097	[0.079	0.116]	0.009	10.202	0.000
DC - > INS - > ASS	−0.012	−0.012	[-0.02	−0.006]	0.003	3.45	0.001
DC - > ASS - > ICA	0.048	0.048	[0.037	0.06]	0.006	8.264	0.000
DE - > DC- > ASS - > ICA	0.018	0.018	[0.014	0.023]	0.002	7.622	0.000
DE - > ASS - > ICA	0.053	0.054	[0.04	0.068]	0.007	7.659	0.000
DE - > DC - > INS - > ASS - > ICA	−0.001	−0.001	[-0.001	0.002]	0.210	3.313	0.001
INS - > ASS - > ICA	−0.012	−0.012	[-0.018	−0.006]	0.003	3.943	0.000
DE - > INS _ - > ASS - > ICA	0.001	0.001	[0.000	0.001]	0.000	0.922	0.357
DE - > DC - > ICA	0.083	0.083	[0.067	0.101]	0.009	9.678	0.000
DC - > INS _ - > ICA	0.044	0.044	[0.034	0.054]	0.005	8.49	0.000
DE - > DC - > INS - > ICA	0.016	0.016	[0.013	0.021]	0.002	7.848	0.000
DE - > INS _ - > ICA	−0.005	−0.005	[-0.014	0.004]	0.005	0.997	0.319
DE - > DC - > INS	0.07	0.07	[0.055	0.086]	0.008	8.74	0.000

M = mean, STD = standard deviation, DE = digital environment, DC = digital competence, INS = instructional practice, ASS=Assessment literacy, ICA-inclusive assessment practice.

5. Discussion of findings

The findings of the study suggest that the impact of the digital environment on fostering inclusive classroom environments is substantial, driven by its provision of tools and resources that cater to diverse student needs. Digital technologies offer customizable interfaces, multimedia content, and assistive technologies, enhancing accessibility for students with disabilities (Florian & Spratt, 2013). For instance, text-to-speech software and interactive visuals enable better engagement with learning materials. Moreover, digital platforms support differentiated instruction, allowing teachers to tailor lessons to various learning styles and abilities, thereby promoting equity in education (Brown, 2016). Research confirms that digital tools improve participation and achievement among students with special educational needs (Smith & Basham, 2014), fostering collaborative learning and community building.

Teachers' digital competence, essential for effectively utilizing digital tools in education, is bolstered by continuous interaction with digital environments. Exposure to these tools enhances teachers' technological pedagogical content knowledge (TPACK), facilitating seamless integration of technology into teaching practices (Tondeur et al., 2017). Professional development through ICT training programs also enhances teachers' confidence and proficiency in utilizing educational software and digital assessment tools (Instefjord & Munthe, 2017), enabling better support for diverse learning needs.

The study findings also suggest that digital environment significantly enhances teachers' assessment literacy by offering sophisticated assessment methods that provide real-time feedback and comprehensive data analysis (Popham, 2010). This capability improves teachers' ability to evaluate student learning effectively and tailor instruction to address individual needs. Immediate feedback from digital assessments aids in identifying learning gaps and implementing timely interventions, thereby enhancing educational outcomes. However, the integration of digital tools does not inherently transform teachers' instructional practices without appropriate pedagogical strategies and professional development (Ertmer & Ottenbreit-Leftwich, 2010). Effective teaching with technology requires alignment with educational goals and student needs, necessitating comprehensive training and pedagogical support.

Digital competences significantly enhance teachers' assessment literacy by equipping them with the skills to effectively design, administer, and interpret assessments using digital tools. Assessment literacy encompasses the understanding of assessment principles and methods crucial for evaluating student learning and guiding instructional decisions. According to research by Tondeur et al. (2017), teachers proficient in digital competences are better prepared to integrate technology into their assessment practices. This integration facilitates the creation of diverse assessment formats such as online quizzes, digital portfolios, and interactive simulations, offering students more authentic and engaging assessment experiences.

The direct influence of instructional practices on teachers' assessment literacy underscores the critical link between teaching methodologies and effective assessment of student learning. Instructional practices encompass the strategies and approaches teachers employ to foster student engagement and understanding. As suggested by Brown (2016), educators who utilize varied and differentiated instructional methods tend to develop robust assessment literacy. This includes the capacity to design assessments aligned with instructional objectives, accurately gauge student comprehension, and provide constructive feedback. Effective instructional practices also support formative assessment, enabling teachers to observe student progress and adjust teaching strategies accordingly. Moreover, Bennett (2011) emphasizes that interactive and student-centered instructional approaches enhance the validity and reliability of assessments, enabling teachers to authentically measure learning outcomes across diverse student groups. Methods like project-based learning and collaborative activities not only enhance engagement but also yield comprehensive data for assessing complex skills and competencies.

Lastly, the direct influence of assessment literacy on inclusive assessment practices underscores the pivotal role of teachers' proficiency in assessment principles in creating fair and accessible assessment environments for all students. Assessment literacy encompasses the knowledge and skills necessary for designing valid, reliable, and fair assessments, essential components of inclusive assessment practices. According to Popham (2010), educators with strong assessment literacy can adeptly adapt assessment methods to meet diverse learning needs and gain valuable insights into student learning progress. Additionally, Stiggins (2002) argues that assessment literacy empowers teachers to implement formative assessment practices crucial for inclusive education, including ongoing assessment and feedback to monitor student progress and tailor instruction to individual student needs.

The significant mediating role of digital competencies between the digital environment and inclusive classroom assessment practices highlights how teachers' digital skills are crucial for promoting inclusive education through effective assessment strategies. Florian and Spratt (2013) argue that digital environments support inclusive practices by providing tools that cater to diverse learning needs, enabling teachers to design accessible assessments, including those for students with disabilities. Tondeur et al. (2017) affirm that teachers with higher digital competencies can effectively utilize digital tools to implement personalized assessment methods and provide adaptable assessments that cater to various learning styles and abilities. For example, digital platforms facilitate alternative assessment formats like multimedia projects or online portfolios, allowing students to demonstrate understanding in diverse ways.

The positive mediation effect of assessment practices between the digital environment and inclusive classroom assessment practices underscores how integrating digital tools enhances educational settings. Assessment practices serve as a vital link, enabling educators to utilize the digital environment for creating inclusive assessments that cater to diverse student needs and promote equitable learning outcomes. Tondeur et al. (2017) indicate that teachers' proficiency in digital competencies enhances their ability to use digital tools effectively in assessments, accommodating different learning styles and abilities. Black and Wiliam (1998) stress that formative assessment practices supported by digital technologies facilitate ongoing feedback and assessment data analysis, essential for adjusting instruction and supporting student progress, fostering continuous improvement in teaching and learning.

The significant mediation effect of instructional practices between digital competence and inclusive classroom assessment practices highlights teachers' instructional strategies' critical role in utilizing digital competences for equitable assessment practices. Ertmer and Ottenbreit-Leftwich (2010) suggest that teachers with strong digital competencies are more likely to employ innovative instructional methods that enhance engagement and promote personalized learning experiences. Combining these practices with digital tools facilitates the design of inclusive assessments that accommodate diverse learner needs and preferences. Effective instructional practices also enable teachers to leverage digital technologies to create accessible assessment formats, provide timely feedback, and differentiate instruction based on individual student responses and needs (Tondeur et al., 2017).

The positive combined impact of digital competence and assessment literacy on inclusive assessment practices highlights the synergistic relationship between teachers' technological proficiency and their capacity to develop inclusive assessments. Digital competence encompasses proficiency in digital tools, while assessment literacy involves knowledge and skills in creating effective assessments. According to Tondeur et al. (2017), teachers proficient in digital tools are more likely to integrate them into assessments, enhancing flexibility and accessibility for diverse students. Similarly, Popham (2010) underscores that assessment literacy enables teachers to design fair and reliable assessments, ensuring equitable evaluation. Guskey (2007) adds that when

digital competence and assessment literacy combine, they support formative assessment practices, fostering personalized instruction and timely feedback using digital technologies.

The significant joint mediation effect of digital competence and instructional practices on inclusive assessment underscores how teachers' technological skills and teaching methods foster inclusive education. Digital competence allows teachers to effectively use digital tools, while instructional practices, as highlighted by Ertmer and Ottenbreit-Leftwich (2010), promote engagement and personalized learning. Integrating these with digital tools enables the creation of adaptable assessments for diverse student needs. Mishra and Koehler (2006) emphasize that Technological Pedagogical Content Knowledge (TPACK) integrates digital competences with pedagogical and content knowledge, enhancing learning opportunities and assessments tailored to varied abilities.

The significant joint mediation effect of instructional practices, digital competence, and assessment literacy on inclusive assessment practices underscores educators' comprehensive role in fostering inclusive educational environments. Instructional practices involve engaging students, while digital competence ensures proficient use of digital tools. Tondeur et al. (2017) note that integrating these with assessment literacy equips teachers to design assessments that cater to diverse needs, leveraging digital tools for varied formats and personalized feedback. Guskey (2007) supports this, suggesting that professional development in instructional strategies and assessment literacy enhances meaningful use of digital technologies in formative assessment, benefiting all students, including those with diverse learning needs.

6. Limitations/suggestion for further studies

The study has several limitations due to its purely cross-sectional design. Firstly, it collects data at a single point in time, which restricts the ability to observe longitudinal changes or trends. This design also cannot establish causality or track how relationships between variables may develop over time. Moreover, the study relies on self-reported data, which is susceptible to biases such as social desirability bias, recall bias, and subjective interpretations by respondents. These factors can undermine the accuracy and reliability of the data collected. Another limitation is that the study's findings are specific to the sampled population. If the sample is not representative of the broader population, the generalizability of the results may be constrained. Variability introduced by differences in participants' backgrounds, including teaching experience, subject area, and prior training in digital tools, could also influence the study's outcomes, and these differences were not fully controlled for. A significant limitation of this study is the potential response bias due to participants' varying digital competencies. Those with higher digital skills might have overestimated their comfort with digital assessment tools, while those with lower competencies might have expressed more dissatisfaction. The study did not address these differences, which could have affected the results.

Future research should assess and adjust for digital competencies to better understand their impact on perceptions of inclusive assessment practices. Furthermore, the study did not fully consider cultural and institutional variations that impact how digital environments and assessment practices are perceived and implemented across different educational contexts. However, these limitations do not render the study findings invalid. Future research could benefit from longitudinal designs to track changes over time and employ more sophisticated methodologies such as mixed methods to gain deeper insights. Additionally, increasing the sample size in future studies would enhance the generalizability of the findings.

7. Conclusions/implications of the study

The study extensively examined the complex interrelationships among digital environment, digital competence, instructional practices, assessment literacy, and inclusive classroom assessment practices. Key findings highlight the direct impacts of the digital environment on digital competence, assessment literacy, and inclusive assessments, emphasizing technology's transformative role in education. Additionally, the study underscores how digital competence directly influences instructional practices and assessment literacy, crucial for equipping teachers with necessary technological skills. It also emphasizes the direct influence of instructional practices on inclusive assessment practices and assessment literacy, underscoring pedagogical strategies' pivotal role in fostering inclusivity.

However, the study found that while instructional practices and assessment literacy individually influence inclusive assessment practices significantly, their combined effect was not statistically significant. Conversely, significant joint mediation effects were observed for digital competence and assessment literacy, as well as digital competence and instructional practices, on inclusive assessment practices. Implications include prioritizing professional development in digital competence and assessment literacy, supported by workshops and continuous training. Educational institutions should establish supportive digital environments providing access to tools and resources essential for fostering teachers' digital competence and implementing inclusive assessments. Facilitating collaborative learning communities among teachers can further enhance sharing of best practices in digital integration and assessment literacy, fostering continuous improvement in inclusive education. Policymakers should implement supportive policies ensuring equitable access to technology and professional development, promoting effective integration of digital tools in educational practices.

Theoretical contributions

The study offers substantial theoretical and practical insights. Theoretically, it presents a detailed framework that clarifies the intricate relationships between the digital environment, digital competence, instructional practices, assessment literacy, and inclusive classroom assessment practices. This enhances existing theoretical models by emphasizing the interconnectedness of these elements. By showing the direct effects of the digital environment on digital competence, assessment literacy, and inclusive assessments, the study strengthens the theoretical concept of technology as a transformative force in education. This supports the incorporation of digital technologies into educational theory and practice. The study also highlights the critical role of digital competence in shaping instructional practices and assessment literacy, underscoring the necessity for educators to develop strong digital skills to effectively employ modern teaching strategies. Furthermore, the significant joint mediation effects of digital competence and assessment literacy, along with digital competence and instructional practices on inclusive assessment practices, provide a sophisticated theoretical understanding of how these factors interact to foster inclusivity. This insight can guide future theoretical models that examine the synergy between technological and pedagogical components in education.

Practical contributions

From a practical standpoint, the study's findings emphasize the importance of integrating technology into all facets of the educational experience. Educational institutions can apply these insights to develop policies and practices that enhance the digital environment, thereby boosting digital competence and assessment literacy among educators. The direct impact of digital competence on instructional practices and assessment literacy indicates that professional development programs should focus on training educators in digital skills. This recommendation can help equip teachers with the necessary technological abilities to improve their teaching effectiveness. By showcasing the influence of instructional practices on inclusive assessment practices, the study offers practical guidance for educators to adopt teaching strategies that support inclusivity, leading to more equitable and effective educational

experiences for all students. The significant joint mediation effects observed suggest that collaborative approaches combining digital competence with instructional practices and assessment literacy can enhance inclusive assessment practices. This practical insight can inform the creation of integrated training programs and collaborative teaching methods. Policymakers can leverage the study's findings to advocate for the inclusion of digital competence and assessment literacy in teacher education curricula, ensuring that future educators are well-prepared to utilize technology in their teaching and assessment practices, ultimately promoting a more inclusive and effective educational environment.

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CRediT authorship contribution statement

Usani Joseph Ofem: Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Eme Orok Iban Amanso:** Writing – review & editing, Project administration, Conceptualization. **Ene I. Ene:** Resources, Project administration, Conceptualization. **Eno Ndaraka Asuquo:** Methodology, Investigation, Conceptualization. **Paulina Unwada Ekpang:** Writing – review & editing, Supervision, Conceptualization. **Joseph Udo Idung:** Writing – review & editing, Writing – original draft, Methodology. **Imelda Barong Edam Agbor:** Writing – review & editing, Validation, Conceptualization. **Eunice Ngozi Ajuluchukwu:** Validation, Resources, Conceptualization. **Annastasia Iwang Anashie:** Writing – original draft, Methodology, Conceptualization. **Eno I. Elogbo:** Validation, Supervision, Conceptualization. **Samson Akinwumi Aderibigbe:** Supervision, Investigation. **Faith Sylvester Orim:** Writing – review & editing, Supervision, Conceptualization. **Stephen Olonye Edeh:** Writing – review & editing, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix 1

Digital environment and competence, instructional practice, assessment literacy and inclusive assessment practices scale (DECIPALIAPS).

S/N	Items	SA	A	D	SD
A	Digital Environment				
1	Our school provides adequate access to digital tools and resources for teaching and learning.				
2	Teachers in our school regularly integrate digital tools into their teaching practices.				
3	Students are taught using online materials in our school.				
4	Digital tools are used for administrative tasks, such as attendance and grading.				
B	Digital competence				
5	I can use digital tools to create and edit digital content				
6	I effectively integrate digital tools into my teaching practices				
7	I am confident in using digital platforms for communication and collaboration with students				
8	I can troubleshoot common technical problems in digital tools.				
9	I regularly update my knowledge about new digital tools and resources				
C	Instructional practices				
10	Students were actively engaged in thought-provoking activities that often involved the critical assessment of procedures.				
11	I ask questions that makes the students to teach during lectures.				
12	I ensured that assignments are given that make students work together.				
13	The lesson encouraged students to explain their reasoning and justify their answers.				
14	Most times, students are allowed to work collaboratively to solve problems during the lesson				
D	Assessment literacy				
15	I can design assessment tasks that align with learning objectives.				
16	I am able to administer assessments in a way that minimizes student anxiety.				
17	I administer assessment that caters for different students needs				
18	I can analyze assessment data to identify student strengths and weaknesses.				
E	Inclusive assessment practices				
19	Assessment methods are designed to accommodate diverse learning styles and needs.				
20	I regularly use various assessment tools (e.g., quizzes, projects, presentations) to cater to different student strengths.				
21	Feedback on assessments is provided in a manner that is understandable and actionable for all students.				
22	I adjust my assessment practices to address the individual needs of students with disabilities.				
	Students are given options to demonstrate their knowledge in different ways (e.g., written, oral, practical).				
23	I incorporate assistive technologies to support students in completing assessments.				

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